MC-PINN: A Feasibility-Oriented Post-Projection PINN for Stiff Heat–Species Coupling

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Abstract

We propose MC-PINN for stiff heat–species coupled reaction–diffusion, where bound violations and target-mean deviations persist under conventional soft penalties. A Fourier-augmented MLP is trained with bound/mean soft losses; at evaluation, species fields undergo alternating box–target-mean projection with a stiffness-inspired time curriculum. Ten-seed experiments cover one-way stiff coupling, bidirectional exothermic coupling, and a mild negative control. On one-way stiff coupling, MC-PINN attains the lowest Rel. $\boldsymbol{L_2}$ with zero violation and near-zero projection residual; on bidirectional stiff coupling, constraint residuals remain near zero; on the mild control, Vanilla PINN is more accurate, indicating that projection is most useful under strong stiffness. We further provide a closure-error bound, heuristic physics-gap diagnostics, and a feasibility–residual Pareto analysis.

Keywords: physics-informed neural networks, MC-PINN, evaluation-time projection, stiff heat–species coupling, physics gap

1 Introduction

Reaction–diffusion systems describe spatiotemporal evolution under diffusive transport and local reaction in population spreading, chemical kinetics, pattern formation, and multiphysics transport[1–4]. Their solutions can exhibit traveling fronts, self-organized patterns, and stiff transients[2–5]: the Fisher-KPP equation models diffusive growth and traveling waves[1,2], the Gray-Scott system generates Turing patterns and serves as a pattern-formation benchmark[3,4], and coupled heat–species problems introduce temporal stiffness and field coupling[5,6]. Traditional solvers use finite differences, finite elements, and IMEX time marching[7,8]. Physics-informed neural networks (PINNs) embed PDE residuals and initial/boundary conditions into the loss, providing a mesh-free route[9,10]; libraries such as DeepXDE have extended PINNs broadly[10,11], Karniadakis et al. survey physics-informed machine learning[11], and benchmarks such as PDEBench emphasize unified comparison of accuracy and physical consistency[12].

On reaction–diffusion problems, PINN limitations have been analyzed through gradient imbalance[13], neural-tangent-kernel theory[14], failure-mode characterization[15], spectral bias[16], and temporal causal stiffness[17]. XPINN extends PINNs by space–time decomposition[18]; Wang et al. add temporal causal weighting to the loss[17]; DeepONet and FNO learn PDE solution operators[19,20]; conservative PINNs enforce conservation by hard projection on discrete domains[21]; neural-operator theory further systematizes PDE solution-map learning on function spaces[22]. Recent work further warns that weak baselines and reporting bias can overstate operator-learning gains[23], and studies causal sampling and second-order optimization

perspectives[24,25]. Most prior studies validate one improvement on one case; how boundary and integral constraints should be chosen, and how loss terms should be configured, remains under-tested across equation types.

Coupled heat–species reaction–diffusion systems can be temporally stiff when fast reaction competes with diffusion: the species field must satisfy physical bounds and an approximate target spatial mean derived from the governing equation, yet conventional PINN soft penalties often yield bound violations or target-mean deviations[9,13,17]. We propose MC-PINN (Multi-Constraint PINN) for periodic stiff heat–species coupling: a Fourier-augmented differentiable neural field with bound/mean soft losses and evaluation-time alternating box–target-mean projection on the species field, plus a stiffness-inspired staged time curriculum. Main experiments cover one-way stiff coupling, bidirectional temperature-dependent exothermic coupling, and a mild negative control; Fisher-KPP, Gray-Scott, and Schnakenberg serve as supplementary applicability cases in Section 3.9. We jointly report L_2, violations, projection residuals, raw/projected PDE residuals, physics gaps, and feasibility–residual trade-offs.

Contributions: (1) a feasibility-oriented post-projection PINN workflow for stiff heat–species coupling with an explicit train-field vs.
projected-field split; (2) first-order mean closure with an error bound, finite-step projection diagnostics, and an operational criterion for projection gaps; (3) ten-seed validation on one-way/bidirectional stiff coupling with quantified gaps, Pareto trade-offs, and inference overhead.

2 Method

2.1 Fourier-augmented neural field

The reaction–diffusion processes discussed in the Introduction are posed on a spatial domain $\Omega \subset \mathbb{R}^{d_s}$ and a time interval $[0, T]$; the unknown field $q(\mathbf{x}, t)$ denotes the physical state (e.g., concentration, temperature, or a multi-species vector). We approximate the continuous spatiotemporal solution with a differentiable neural field $q_\theta(\mathbf{x}, t)$, where θ are network parameters and the output dimension d_q equals the number of state components.

$$q_\theta(\mathbf{x}, t) \in \mathbb{R}^{d_q}, \qquad \mathbf{x} \in \Omega, \quad t \in [0, T]. \tag{2-1}$$

Here $\mathbf{x}$ is the spatial coordinate, t is time, Ω is the spatial domain, T is the final time, and d_q is the output dimension.

Reaction–diffusion solutions often carry traveling fronts, local patterns, and multiscale spatial structures, and approximating q_θ with a plain MLP is easily hampered by low-frequency bias. We use a Fourier-feature-augmented MLP (FF-PINN style) as the backbone: fixed Fourier bases plus a Tanh MLP. The spatiotemporal coordinates are first normalized linearly:

$$\tilde{\mathbf{x}} = \frac{\mathbf{x} - \mathbf{x}_{\min}}{\mathbf{x}_{\max} - \mathbf{x}_{\min}}, \qquad \tilde{t} = \frac{t - t_{\min}}{t_{\max} - t_{\min}}. \tag{2-2}$$

Here $\mathbf{x}_{\min}, \mathbf{x}_{\max}$ and $t_{\min}, t_{\max}$ are normalization endpoints, and $\tilde{\mathbf{x}}, \tilde{t}$ are normalized coordinates.

The Fourier feature mapping uses sine and cosine basis functions, with the general form

$$\gamma(\mathbf{v}) = \left[\cos(2\pi \mathbf{B}^\mathsf{T} \mathbf{v}),\ \sin(2\pi \mathbf{B}^\mathsf{T} \mathbf{v})\right]^\mathsf{T}. \tag{2-3}$$

Here $\mathbf{v}$ is the input vector, $\mathbf{B}$ is the frequency matrix, and $\gamma(\cdot)$ is the Fourier feature map.

In our implementation the frequencies are spaced logarithmically; for each normalized coordinate component $\tilde{x}_d$ we set

$$\omega_k = 2^k \pi, \qquad \gamma_d^{(k)}(\tilde{x}_d) = \left[\sin(\omega_k \tilde{x}_d),\ \cos(\omega_k \tilde{x}_d)\right], \quad k = 0, \ldots, K-1. \tag{2-4}$$

Here $\tilde{x}_d$ is the d-th component of $\tilde{\mathbf{x}}$ or $\tilde{t}$, ω_k is the k-th frequency, and K is the number of frequency levels per coordinate.

Concatenating the original coordinates with the Fourier features yields the network input vector. Here $\gamma(\tilde{\mathbf{x}}, \tilde{t})$ denotes the concatenation of per-component encodings from Eq. (2-4):

$$\mathbf{z} = \left[\mathbf{x},\ t,\ \gamma(\tilde{\mathbf{x}}, \tilde{t})\right]^{\mathsf{T}}. \tag{2-5}$$

Here $\mathbf{z}$ is the MLP input vector.

An L-layer Tanh fully connected network then outputs the neural field:

$$q_\theta(\mathbf{x}, t) = \mathcal{N}_\theta(\mathbf{z}), \qquad \mathcal{N}_\theta : \mathbb{R}^{d_z} \to \mathbb{R}^{d_q}. \tag{2-6}$$

Here $\mathcal{N}_\theta$ is an L-layer Tanh MLP, d_z is the input dimension, and L is the number of hidden layers.

The w/o Fourier ablation removes $\gamma(\cdot)$ and reduces to a vanilla MLP field.

2.2 Physics-aware multi-constraint projection and training objective

The neural field q_θ remains differentiable for data fitting and PDE-residual automatic differentiation; the reported prediction $\hat{q}$ is obtained from q_θ by physics-aware projection. Denoting the raw prediction by q_θ and the projected one by $\hat{q}$,

$$\hat{q} = \Pi(q_\theta). \tag{2-7}$$

Here Π is the projection operator induced by physical constraints, and $\hat{q}$ is the feasible field used at evaluation.

For a scalar component q_i that must lie in $[a, b]$, the box projection reads

$$\hat{q}_i(\mathbf{x}, t) = \min\{b, \max\{a, q_{\theta,i}(\mathbf{x}, t)\}\}. \tag{2-8}$$

Here $q_{\theta,i}$ is the raw network prediction for component i, and a, b are the physical lower and upper bounds. For vector fields with multiple bounded components, Eq. (2-8) is applied component-wise.

On a periodic domain, when the reaction term changes the spatial mean of a species field, a target-mean constraint is needed in addition to box bounds. For species concentration c, integrating the species equation under periodic boundaries makes the spatial integral of the diffusion term vanish, so that

$$\frac{dM_c}{dt} = k_r \langle c(1-c) \rangle, \qquad M_c(t) = \langle c(\cdot, t) \rangle. \tag{2-9}$$

Here $M_c(t)$ is the spatial mean of c at time t, $\langle \cdot \rangle$ denotes spatial averaging over $x \in \Omega$, and k_r is the reaction-rate constant.

Under the assumptions of smooth initial values and weak spatial variance, with the mean-field closure $\langle c(1-c) \rangle \approx M_c(1 - M_c)$, the target mean satisfies

$$\dot{M_c} = k_r M_c (1 - M_c), \qquad M_c^\star(t) = \frac{M_{c0}}{M_{c0} + (1 - M_{c0}) e^{-k_r t}}. \tag{2-10}$$

Here $M_c^\star(t)$ is the target spatial mean from the closure ODE and $M_{c0} = M_c(0)$ is the initial mean.

The temperature-field mean is then given approximately from the reaction heat and the species-mean change:

$$M_T^\star(t) \approx M_{T0} + \alpha \left(M_c^\star(t) - M_{c0} \right). \tag{2-11}$$

Here $M_T^\star(t)$ is the target temperature mean, M_{T0} is the initial temperature mean, and α is the exothermic coupling coefficient.

Writing the network-predicted mean as $m_\theta(t) = \langle c_\theta(\cdot, t)\rangle$, the target-mean projection reads

$$c^{(0)}(x,t) = c_\theta(x,t) + M_c^\star(t) - m_\theta(t). \tag{2-12}$$

Here c_θ is the raw species prediction and $c^{(0)}$ is the mean-corrected intermediate field. The closure $\langle c(1-c)\rangle \approx M_c(1-M_c)$ induces error $\epsilon_{\text{closure}} = -\text{Var}[c]$; on the reference solution, the temporal mean of $|\epsilon_{\text{closure}}|$ is 2.7×10^{-3} and the maximum is 2.0×10^{-2}, which can locally exceed the network L_2 error. Projection on $M_c^\star(t)$ is therefore an approximate-target feasibility correction, not a strict hard constraint on the true solution. Alternating projection between the mean hyperplane and $c \in [0,1]$ follows Algorithm 1 with $N_{\text{iter}} = 60$. Both constraint sets are convex; when $M_c^\star(t) \in [0,1]$ and the intersection is nonempty, von Neumann alternating projection converges to a point in the intersection, but that point is generally not the PDE ground truth.

Algorithm 1 Alternating box–target-mean projection on periodic species fields
Input: network prediction $c_\theta(\cdot,t)$, target mean $M_c^\star(t)$, iteration count N_{iter}
Output: feasible $\hat{c}(\cdot,t) \in [0,1]$ whose finite-iteration mean approximately matches $M_c^\star(t)$
1. Initialize $c^{(0)}(x,t) = c_\theta(x,t) + M_c^\star(t) - \langle c_\theta(\cdot,t)\rangle$;
2. **for** $k = 1,\ldots,N_{\text{iter}}$ **do**
2.1. Mean projection: $c^{(k-\frac{1}{2})}(x,t) = c^{(k-1)}(x,t) + M_c^\star(t) - \langle c^{(k-1)}(\cdot,t)\rangle$;
2.2. Box constraint: $c^{(k)}(x,t) = \min\{1, \max\{0, c^{(k-\frac{1}{2})}(x,t)\}\}$;
3. **end for**; return $\hat{c} = c^{(N_{\text{iter}})}$.

Fig. 1: Alternating box–target-mean projection on periodic species fields.

The training objective is a weighted sum of data, initial/boundary, PDE-residual, snapshot, and bound/mean constraint terms:

$$\mathcal{L}(\theta) = \lambda_d\mathcal{L}_{data} + \lambda_{ic}\mathcal{L}_{ic} + \lambda_{bc}\mathcal{L}_{bc} + \lambda_r\mathcal{L}_{res} + \lambda_s\mathcal{L}_{snap} + \lambda_b\mathcal{L}_{bound} + \lambda_m\mathcal{L}_{mean}. \tag{2-13}$$

Here $\mathcal{L}_{data}, \mathcal{L}_{ic}, \mathcal{L}_{bc}, \mathcal{L}_{res}, \mathcal{L}_{snap}$ are data, initial, boundary, PDE-residual, and snapshot losses; $\mathcal{L}_{bound}$ penalizes bound violations of the raw output and $\mathcal{L}_{mean}$ penalizes the raw mean against $M_c^\star(t)$; $\lambda_\cdot$ are the corresponding weights.

The PDE residual is computed by automatic differentiation on the differentiable q_θ, not on projected $\hat{q}$; training optimizes the differentiable field and evaluation uses the projected $\hat{q}$. For a general reaction–diffusion system $\partial_t q = \nabla\cdot(\mathbf{D}\nabla q) + R(q)$, the residual is

$$\mathcal{R}_\theta(\mathbf{x},t) = \partial_t q_\theta - \nabla\cdot(\mathbf{D}\nabla q_\theta) - R(q_\theta), \tag{2-14}$$

Here $\mathbf{D}$ is the diffusion matrix (diagonal in our experiments), $R(q)$ is the reaction term, and $\mathcal{R}_\theta$ is the discrete PDE residual.

The residual loss is

$$\mathcal{L}_{res} = \frac{1}{N_r}\sum_{i=1}^{N_r} \|\mathcal{R}_\theta(\mathbf{x}_i, t_i)\|_2^2. \tag{2-15}$$

Here N_r is the number of collocation points and $(\mathbf{x}_i, t_i)$ is the i-th collocation coordinate.

2.3 Stiffness-informed staged time-window curriculum

Training uses a preset expanding time-window curriculum. Stage s ends at $T_s = \rho_s T$ with $0 < \rho_1 < \cdots < \rho_S = 1$; samples are drawn on $\Omega \times [0, T_s]$ with loss weight β_s. Fractions $\{\rho_s\}$ and weights $\{\beta_s\}$ are set from equation time scales: reaction time scale $\tau_r = 1/k_r$ and Damköhler number $Da = k_r T$ suggest shorter early

windows with larger weights before expanding to the full horizon. Specific stage settings are given in the experiments section.

$$T_s = \rho_s T, \qquad 0 < \rho_1 < \rho_2 < \cdots < \rho_S = 1. \tag{2-16}$$

Here S is the number of curriculum stages, ρ_s is the time-window fraction at stage s, and β_s is the stage loss weight.

2.4 MC-PINN formulation for stiff heat–species coupling

On a periodic domain, the 1D coupled system reads

$$\partial_t \mathbf{q} = \nabla \cdot (\mathbf{D} \nabla \mathbf{q}) + \mathbf{R}(\mathbf{q}), \qquad \mathbf{q} = (T, c)^{\mathsf{T}}, \qquad \mathbf{q} : \Omega \times [0, T] \to \mathbb{R}^2. \tag{2-17}$$

Here $\mathbf{q}$ is the state vector, T is temperature, c is species concentration, $\mathbf{D}$ is the diffusion matrix, and $\mathbf{R}(\mathbf{q})$ is the coupled reaction source.

MC-PINN separates the differentiable training field from the projected evaluation field:

$$\mathbf{q}_\theta(\mathbf{x}, t) = \mathcal{N}_\theta\big([\mathbf{x}, t, \gamma(\tilde{\mathbf{x}}, \tilde{t})]\big), \qquad \hat{\mathbf{q}}(\mathbf{x}, t) = \Pi_{\mathcal{C}}(\mathbf{q}_\theta(\mathbf{x}, t)), \tag{2-18}$$

Here $\mathbf{q}_\theta$ is the differentiable training field, $\hat{\mathbf{q}}$ is the projected evaluation field, and $\Pi_{\mathcal{C}}$ is the alternating box–target-mean projection on species c; temperature T receives bound projection only at evaluation. Eq. (2-18) keeps non-smooth clipping out of second-order AD; closure-error bounds and projection-induced physics gaps are analyzed below.

2.5 Projection gap, closure error, and feasibility criteria

Projection $\mathcal{P}$ enforces feasibility but generally does not preserve the PDE residual. With raw prediction q_θ and $\hat{q} = \mathcal{P}(q_\theta)$,

$$\mathcal{R}(\hat{q}) = \mathcal{R}(q_\theta) + \underbrace{\big(\mathcal{R}(\mathcal{P}(q_\theta)) - \mathcal{R}(q_\theta)\big)}_{\text{physics gap}}, \tag{2-19}$$

Here q_θ is the raw neural prediction, $\hat{q} = \mathcal{P}(q_\theta)$ is the projected field, $\mathcal{R}(\cdot)$ is the discrete PDE residual, and $\mathcal{P}$ is the evaluation-time projection operator. Eq. (2-19) separates the post-projection residual change as a physics gap; later sections use it to diagnose how evaluation-time projection affects PDE residuals.

Closure-error bound. The mean closure $\langle c(1-c)\rangle \approx M_c(1-M_c)$ has error $\epsilon_{\text{closure}} = -\text{Var}[c]$. For $c \in [0,1]$, $\text{Var}[c] \le 1/4$ hence $|\epsilon_{\text{closure}}| \le 0.25$; on our reference solution the temporal mean of $|\epsilon_{\text{closure}}|$ is 2.7e-3, 95% percentile 0.019, maximum 0.02, locally comparable to network L_2. Thus $M_c^\star(t)$ is an approximate target and projection corrects feasibility rather than the ground truth.

Bidirectional exothermic closure ODE (heuristic). Under $\langle k(T)c(1-c)\rangle \approx k(M_T)M_c(1-M_c)$,

$$\frac{\mathrm{d}M_c}{\mathrm{d}t} = k(M_T)\, M_c(1-M_c), \qquad \frac{\mathrm{d}M_T}{\mathrm{d}t} = \alpha\, k(M_T)\, M_c(1-M_c), \tag{2-20}$$

Here M_c, M_T are the spatial means of species and temperature, $k(M_T)$ is the temperature-dependent reaction rate, and α is the exothermic coupling coefficient. For one-way coupling, $k(M_T) \equiv k_r$ (Eq. (2-9)); for bidirectional stiff coupling, $k(M_T) = k_r \exp(\beta(M_T - T_{\text{ref}}))$.

Finite-step projection diagnostics. Both $C = [0,1]^{N_x}$ and the mean hyperplane H_{M_c} are convex; von Neumann alternating projection converges to $C \cap H_{M_c}$ when nonempty, but that point need not be the PDE solution. For w/o MC predictions with $N_{\text{iter}} \in \{8, 60\}$, the target-mean residual decays monotonically while Rel. L_2 is nearly unchanged: projection mainly adjusts feasibility, not pointwise accuracy.

Operational criterion. Use evaluation-time projection when (i) violation $\le \varepsilon_v$ (we take 10^{-4}), (ii) projection residual $\le \varepsilon_m$ (10^{-5}), and (iii) physics gap $\le \kappa L_2$ (on one-way stiff coupling we take $\kappa = 30$,

matching an MC-PINN gap-to-L_2 ratio of roughly $0.31/9.8e-3$); otherwise prefer non-projected PINNs or improved closure/training. Later sections visualize the violation–residual–gap trade-off as a Pareto plot.

3 Experiments

3.1 Coupled governing equations and main validation cases

The main text focuses on three periodic 1D heat–species coupled systems: one-way stiff coupling, a mild negative control, and bidirectional exothermic stiff coupling.

(1) One-way stiff coupling. On $x \in [0,1]$, $t \in [0,0.5]$, with periodic boundaries,

$$\begin{aligned} T_t &= D_T\, T_{xx} + \alpha k_r\, c(1-c), \\ c_t &= D_c\, c_{xx} + k_r\, c(1-c), \end{aligned} \qquad D_T = 0.025,\ D_c = 0.003,\ k_r = 25,\ \alpha = 1.2. \tag{3-1}$$

Here T, c are temperature and species concentration, D_T, D_c are diffusion coefficients, k_r is the reaction rate, α is the exothermic coupling coefficient, and L is the domain length.

Initial data $c(x,0) = 0.45 + 0.20\cos(2\pi x/L)$, $T(x,0) = 1.0 + 0.1\sin(2\pi x/L)$. The species mean is autonomous on the periodic domain; $M_c^\star(t)$ is derived and enforced by the alternating projection in Section 2.2.

(2) Mild negative control. Same form as Eq. (3-1) with $k_r = 1$, $D_c = 0.02$, $D_T = 0.08$, $\alpha = 0.2$.

(3) Bidirectional stiff coupling. Reaction rate $k(T) = k_r \exp(\beta(T - T_{\text{ref}}))$ ($\beta = 1.5$) with exothermic feedback $\alpha k(T) c(1-c)$; mean targets follow the closure ODE in Section 2.5.

Reference solutions use 1D FD with CFL subcycling[7,8]; non-standard finite-difference models for reaction–diffusion equations can preserve positivity and boundedness[26]. Metrics: Rel. L_2, violations, projection residuals, raw/projected PDE p_{95}, and physics gaps.

Table 1 summarizes the three main validation cases: state variables, spatiotemporal domains, boundary conditions, and reference-solution generation.

Table 1: Main-text heat–species coupling cases.

Case	State variable	Spatial domain	Time domain	Boundary	Reference solution
One-way stiff	(T, c)	$[0,1]$	$[0,0.5]$	periodic	Finite-difference (stiff subcycling)
Mild (negative)	(T, c)	$[0,1]$	$[0,0.5]$	periodic	Finite-difference (mild stiffness)
Bidirectional stiff	(T, c)	$[0,1]$	$[0,0.5]$	periodic	Finite-difference (bidirectional exothermic stiff)

3.2 Experimental setup, hyperparameters, and baselines

This subsection reports the ten-seed protocol, shared coupled-case hyperparameters, and baseline differences.

Table 2 summarizes hyperparameters shared by the three coupling cases.

Table 3 contrasts MC-PINN with literature baselines (main text focuses on coupling cases).

One-way/bidirectional stiff coupling uses 4 layers / 128 hidden units, Fourier $n_{\text{freq}} = 6$, six-stage curriculum (3500 epochs); the mild negative control uses 2800 epochs.

Vanilla and Fourier-PINN use MLP soft penalties with uniform sampling; Causal-PINN shares the staged window but adds Wang et al.[17] causal weighting without physics projection; PINN-Proj enforces

Table 2: Coupled-case hyperparameters (shared except backbone and projection/curriculum switches)

Item	One-way / bidirectional stiff	Mild negative control
Layers / hidden width	4 / 128	4 / 128
Fourier n_{freq}	6	6
Training points	12000	12000
Collocation points	6000	6000
PDE / snapshot weights	0.12 / 0.45	0.12 / 0.45
Species-channel loss weight	3.0	3.0
Stage count	6	6
Epochs	3500	2800

Table 3: Method difference matrix

Method	Neural field	Constraint	Time strategy	Prior RD tests
Vanilla PINN[9]	MLP + Tanh	Soft penalty	Uniform sampling	Burgers / Allen–Cahn
Fourier-PINN[16]	Fourier-augmented MLP	Soft penalty	Uniform sampling	Fisher/Gray-Scott herein
PINN-Proj[21]	MLP	Hard mean/conservation projection	Uniform sampling	Linear conservation
Causal PINN[17]	MLP	Soft penalty	Staged*+causal bins	Allen–Cahn / KdV
FNO[20,22]	Spectral conv. operator	No explicit constraint	Offline grid supervision	Burgers / Darcy
MC-PINN (this work)	Fourier-augmented MLP	Bound / target-mean losses + prediction projection	Stiffness-inspired staged window	Coupled heat–species main cases

*Vanilla and Fourier-PINN use uniform full-horizon sampling; Causal-PINN and MC-PINN share the staged fractions in Table 2 on each case. Causal-PINN additionally applies Wang et al.[17] residual time-bin causal weighting and has no physics projection.

conservation by hard projection[21]; MC-PINN combines a Fourier field, equation-specific constraint losses, evaluation-time projection, and staged windows.

Each (method, case) pair uses 10 fixed seeds (20260523–20261001); the main text reports one-way/bidirectional stiff coupling and the mild negative control with mean ± std and 95% CIs. Six literature baselines include Causal-PINN (fair) (causal weighting only). All methods share AdamW (lr=2×10^{-3}, weight_decay=10^{-5}), Tanh activation, and the sampling budget in Table 2. Experiments ran on an NVIDIA RTX 4060 Ti; GPU-batched projection adds millisecond-level overhead. Ablations remove the MC projection module, Fourier features, or the time curriculum.

3.3 One-way stiff coupled heat–species system

Among six main baselines, MC-PINN attains the lowest Rel. L_2 (9.8e-3 ± 1.7e-3); Wilcoxon $p = 0.002$, Cohen's $d = 1.06$, lower in 10/10. Vanilla PINN 0.027 ± 0.017. MC-PINN projection residual 4.9e-6, zero violation; Vanilla/Fourier violations 0.11 / 0.096.

Table 4 lists full-field Rel. L_2 errors from independent training on each fixed seed.

MC-PINN is lower than Vanilla PINN on all ten seeds; Vanilla shows its largest outlier on 20260701 (0.073) while MC-PINN stays at 0.0137 on the same seed. Fourier-PINN varies less seed-to-seed but remains above MC-PINN on most entries.

Figure 2 shows the spatiotemporal T, c structure for one-way stiff coupling. The top row shows reaction heating raising the temperature plateau over time; the bottom row preserves periodic c stripes with aligned

Table 4: Per-seed Rel. L_2 on the stiff coupled case (six main baselines, ten seeds).

Seed	Vanilla PINN	Fourier-PINN	PINN-Proj	Causal-PINN	Causal-PINN (fair)	MC-PINN
20260523	0.0182	8.7e-3	0.0369	0.0156	4.39e-3	8.52e-3
20260601	0.0307	9.3e-3	0.0166	0.0141	7.24e-3	0.0115
20260618	0.0123	0.0106	0.0104	0.0626	0.0159	8.68e-3
20260701	0.073	0.0101	0.0272	0.0323	9.94e-3	0.0137
20260715	0.025	0.0107	9.56e-3	0.0106	0.0104	9.45e-3
20260801	0.0288	0.0109	0.0305	0.0121	0.0148	8.32e-3
20260815	0.0182	9.73e-3	0.0321	0.0135	0.0151	0.0111
20260901	0.0233	0.0101	9.68e-3	0.012	0.0131	8.21e-3
20260915	0.0121	0.0116	0.012	0.017	8.18e-3	9.2e-3
20261001	0.0276	0.0137	0.0299	0.012	0.0145	9e-3

peaks/troughs. Centerline slices show largest T errors in the reactive band ($x \approx 0.3$–0.7) and Δc concentrated near stripe boundaries at about 10^{-2}.

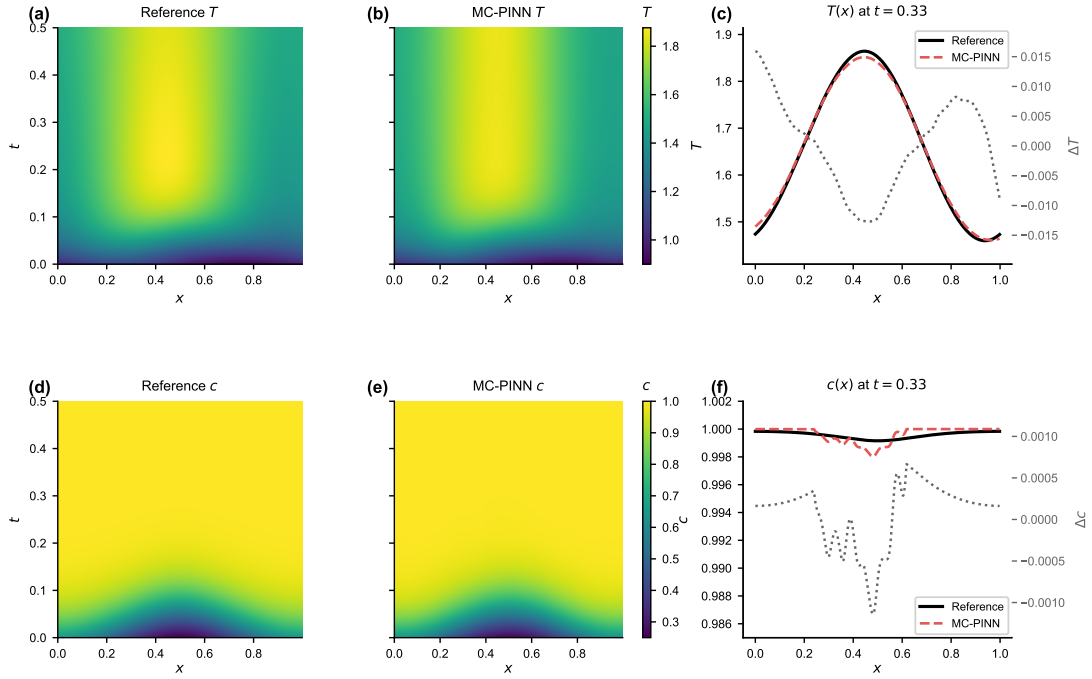


Fig. 2: One-way stiff coupling ($k_r = 25$): reference and MC-PINN T, c fields (t–x), centerline slices at $t \approx 0.33$ (solid: reference, dashed: MC-PINN), and $\Delta T, \Delta c$ on the right axis (dotted).

Table 5 quantifies how evaluation-time projection changes PDE residuals; for most methods $\mathcal{R}_{proj}$ exceeds $\mathcal{R}_{raw}$, highlighting a tension between feasibility correction and discrete PDE consistency.

PINN-Proj lowers $\mathcal{R}_{proj}$ (0.28) slightly below $\mathcal{R}_{raw}$ (0.31); MC-PINN has the lowest $\mathcal{R}_{raw}$ (0.17) but $\mathcal{R}_{proj}$ rises to 0.48 with gap 0.31, showing projection can raise discrete residuals while improving feasibility. Full MC-PINN has a slightly lower gap than w/o curriculum (0.35).

Figure 3 tracks deviation of the spatial species mean from $M^\star(t)$. Non-projected baselines drift early; PINN-Proj and MC-PINN keep the mean-projection residual near 10^{-6} for $t \geq 0.3$. This metric differs from the projected PDE residual $\mathcal{R}_{proj}$ in Table 5.

Table 5: Stiff coupled: raw/projected PDE residual and physics gap (ten-seed mean). $\mathcal{R}_{raw}$ and $\mathcal{R}_{proj}$ are mean p_{95} residuals before/after projection; the gap column is the mean over seeds of $|p_{95}(\mathcal{R}(\hat{q})) - p_{95}(\mathcal{R}(q_\theta))|$.

Method	$\mathcal{R}_{raw}$	$\mathcal{R}_{proj}$	Proj.-induced gap
PINN-Proj	0.31	0.28	0.26
MC-PINN	0.17	0.48	0.31
w/o Fourier	0.22	0.31	0.32
w/o curriculum	0.15	0.5	0.35



Fig. 3: Species-mean projection residual vs. $M^\star(t)$. (a) Full-horizon comparison of five methods; non-projected baselines (Vanilla/Fourier/Causal) drift from the feasible band. (b) Zoom for $t \geq 0.3$; PINN-Proj and MC-PINN stay near 10^{-6}.

3.4 Bidirectional stiff coupled heat–species system

With $k(T) = k_r \exp(\beta(T - T_{\text{ref}}))$ and exothermic feedback, we test whether bidirectional coupling changes the feasibility–accuracy trade-off under the same ten-seed protocol.

Table 6 summarizes ten-seed mean Rel. L_2, projection residual, and violation under bidirectional exothermic feedback.

Table 6: Bidirectional stiff coupling: $k(T) = k_r e^{\beta(T-T_{\text{ref}})}$; 10-seed Rel. L_2 / mean residual / violation

Method	Rel. L_2	Mean residual	Violation
Vanilla PINN	0.25 ± 0.19	0.69	0.017
Fourier-PINN	0.019 ± 5.5e-3	0.051	0.27
PINN-Proj	0.059 ± 0.023	8.1e-6	0
Causal-PINN	0.065 ± 0.048	0.21	0.31
Causal-PINN (fair)	0.098 ± 0.011	0.49	0.11
MC-PINN	0.058 ± 0.027	5.5e-6	0

The rate $k(T)c(1-c)$ drives both species and heat equations; mean targets follow the closure ODE (Section 2.5).

MC-PINN Rel. L_2 0.058 ± 0.027, mean residual 5.5e-6, violation 0; Vanilla PINN Rel. L_2 0.25 ± 0.19. Paired t-test $p = 0.010$ (lower/total 10/10). Fourier-PINN can be lower in L_2 but with violation 0.27, showing that minimizing L_2 alone does not ensure feasibility.

Figure 4 shows bidirectional T, c evolution. Temperature rises more strongly than in the one-way case.

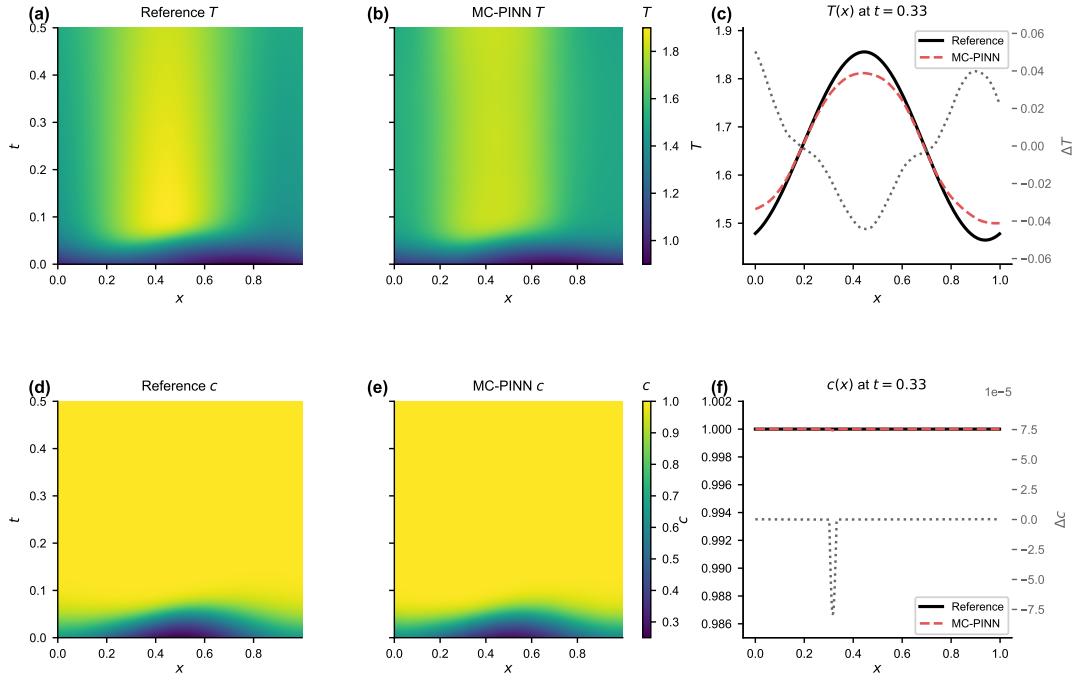


Fig. 4: Bidirectional stiff coupling ($k(T) = k_r e^{\beta(T-T_{\text{ref}})}$): T, c fields and centerline slices.

3.5 Feasibility–residual Pareto and projection gap

Section 2.5 criteria (i)–(iii) require joint inspection of violation and projected PDE residuals; the next figure plots their Pareto trade-off on one-way and bidirectional stiff coupling.

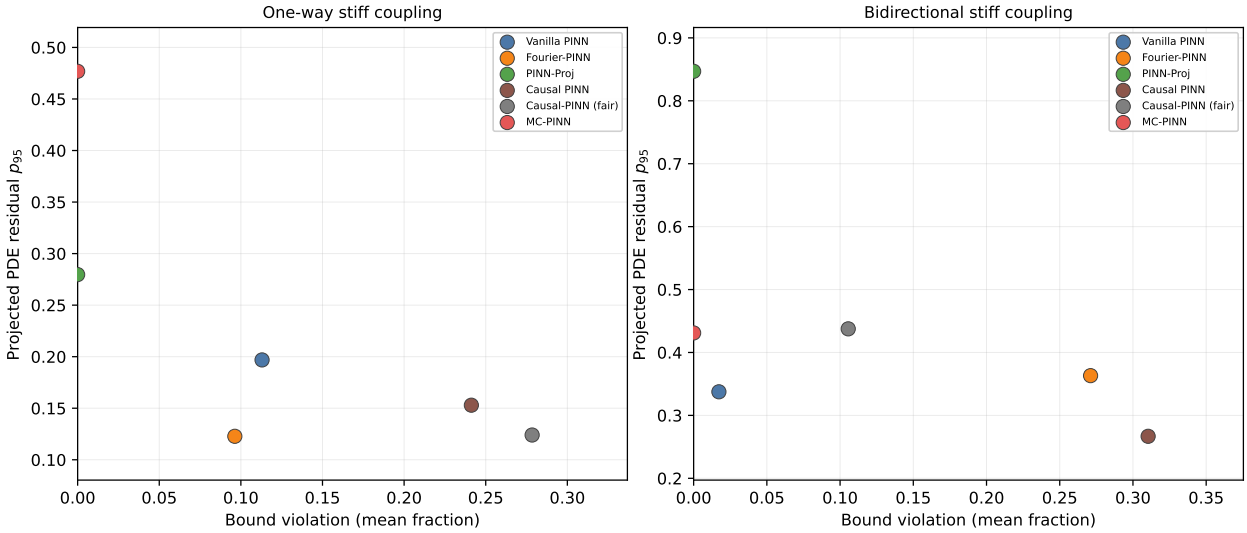


Fig. 5: Feasibility–residual Pareto: violation vs. projected PDE p_{95} (ten-seed means). Low-violation methods (MC-PINN/PINN-Proj) tend to have higher projected residuals.

Figure 5 (left): on one-way stiff coupling MC-PINN/PINN-Proj sit in the low-violation region with higher projected residuals; (right): bidirectional case shows the same trade-off—Fourier-PINN can be lower in L_2 but with large violation (Table 6). Together with Section 2.5 criteria (i)–(iii), this shows projection trades feasibility for global PDE consistency.

3.6 Summary and ablations

This subsection summarizes Rel. L_2 on the three coupling cases, stiff-coupled ablations, and paired t-tests versus MC-PINN. On the mild negative control ($k_r = 1$), Vanilla PINN is lowest in L_2 (7.9e-3 $\pm$ 2.6e-3), MC-PINN 0.021 $\pm$ 1.3e-3—projection is most useful under strong stiffness.

Figure 6 compares three coupling regimes. MC-PINN is lowest with small variance on one-way stiff coupling; Vanilla PINN wins on the mild negative control ($k_r = 1$); on bidirectional stiff coupling Vanilla has the largest spread and Fourier-PINN is lowest in mean but with large violation.

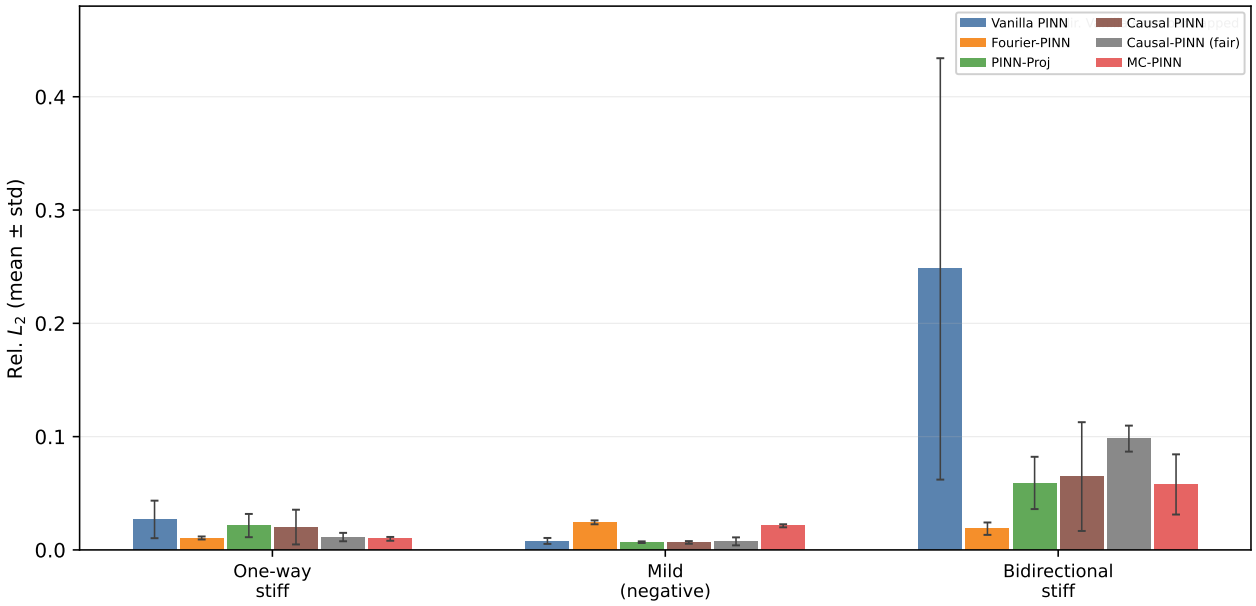


Fig. 6: Rel. L_2 on three coupling cases (ten-seed mean ± std). MC-PINN is lowest on one-way stiff coupling; Vanilla PINN is lowest on the mild negative control; on bidirectional stiff coupling Vanilla has the largest spread and Fourier-PINN can be lowest in mean but with large violation.

Figure 7 ablations show w/o MC raises projection residual (0.29) and violation; w/o Fourier raises both L_2 and physics gap; w/o curriculum can slightly lower L_2 (7.2e-3) but increases violation, indicating curriculum and projection act complementarily on feasibility vs. accuracy.

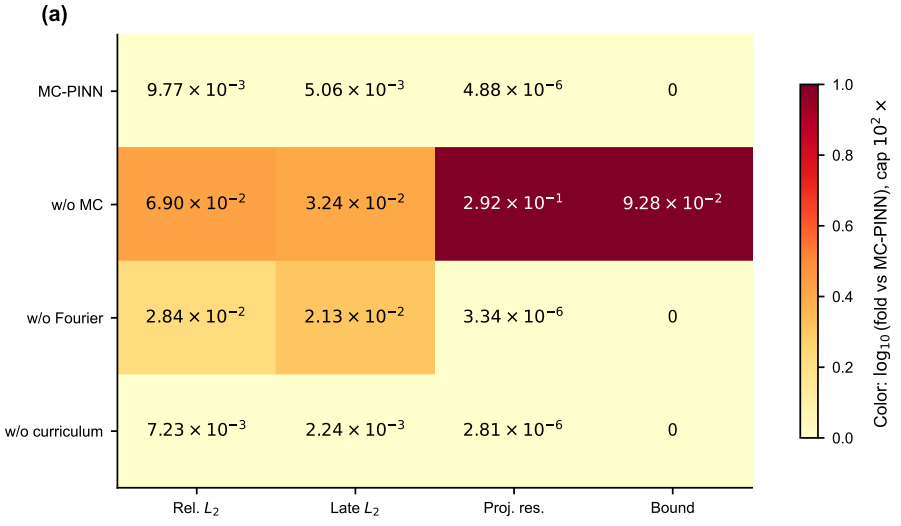


Fig. 7: Stiff-coupled ablation heatmap: w/o MC raises projection residual and violation; w/o Fourier raises L_2 and physics gap; w/o curriculum lowers L_2 but increases violation.

Table 7 reports paired t-tests of MC-PINN versus Vanilla PINN on coupling cases with Bonferroni correction.

One-way stiff coupling: $p = 0.009$ (not significant after Bonferroni); mild negative control: $p < 10^{-4}$ but MC-PINN is lower in 0/10 seeds. Only coupling cases are listed. Wilcoxon signed-rank (stiff vs. Vanilla):

Table 7: Paired t-tests of MC-PINN vs main baselines (10 seeds, L_2 error)

Case	Baseline	Mean diff.	95% CI	p	Lower/total
Stiff coupled	Vanilla PINN	-0.017	±0.012	0.009	10/10
Mild coupled	Vanilla PINN	0.013	±2.6e-3	0.000	0/10

$p = 0.002$, Cohen's $d = 1.06$, lower/total 10/10. Bonferroni-corrected p within family: stiff $p_{\mathrm{B}} = 0.255$; mild $p_{\mathrm{B}} = 0.000e + 00$.

3.7 Strong baselines and noise robustness

Adding 1%/5% relative IC noise (Figure 8), MC-PINN keeps violation at zero while Vanilla violation rises with noise, showing robust feasibility correction at evaluation.

Table 8 compares Sigmoid hard bounds (HardBound), strong penalties (StrongPenalty), and augmented Lagrangian (AugLag): HardBound suppresses violation but raises L_2; StrongPenalty/AugLag remain non-feasible and do not beat MC-PINN in L_2, supporting soft training plus evaluation-time projection.

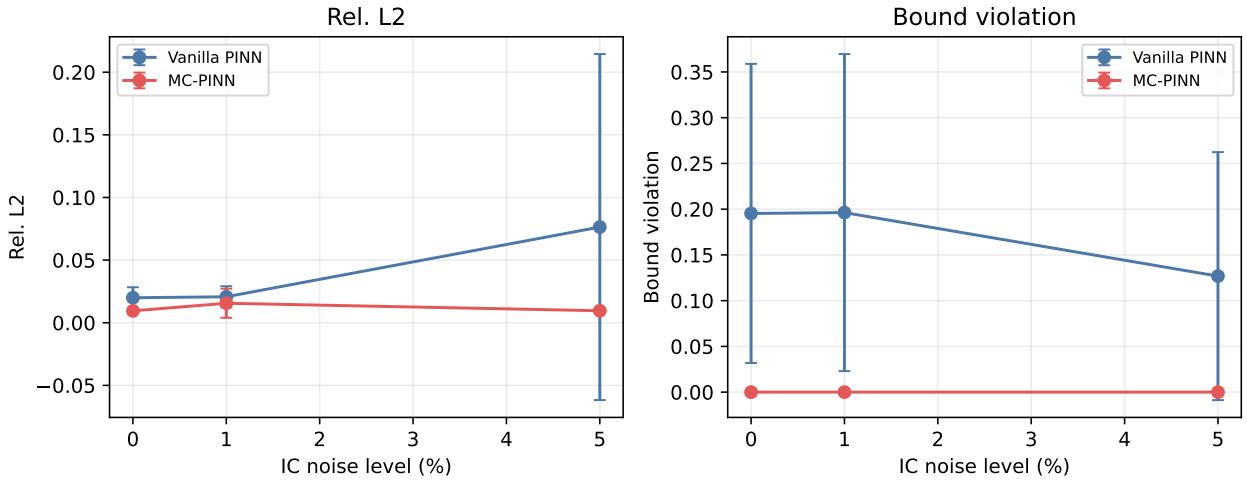


Fig. 8: Rel. L_2 and violation under 0/1/5% IC noise.

Table 8: Strong-constraint baselines (one-way stiff coupling, ten seeds)

Method	Rel. L_2	Violation	Proj. residual
HardBound	0.25 ± 4.8e-4	0	0.5
StrongPenalty	0.2 ± 0.015	0.33	0.43
AugLag	0.15 ± 0.017	0.32	0.38
MC-PINN	9.5e-3 ± 1.1e-3	0	4.7e-6

3.8 Applicability control experiments

The following cases are not ranked against MC-PINN in the main text; they illustrate applicability boundaries for bound projection, snapshot supervision, and settings without a mean constraint.

Fisher-KPP. Ten-seed Rel. L_2: Vanilla PINN 3e-3 ± 3.5e-4, MC-PINN 8e-3 ± 0.016, paired $p = 0.35$ (not significant). MC-PINN mainly keeps violation at zero.

Gray-Scott. Fourier-PINN lowest 8.3e-3 ± 3.8e-4, MC-PINN 0.011 ± 6.7e-3. Main runs use $\lambda_r = 0$ (snapshot-driven).

Schnakenberg. Fourier-PINN lowest 9.8e-3 ± 1.8e-4, MC-PINN 0.011 ± 2.9e-3, paired $p = 0.26$ (not significant).

Table 9 summarizes Rel. L_2 and violation on the three applicability cases (not ranked against MC-PINN in the main narrative).

Table 9: Applicability controls: Fisher-KPP, Gray-Scott, and Schnakenberg (ten-seed mean ± std Rel. L_2)

Case	Lowest-L_2 baseline A	Lowest-L_2 baseline B	MC-PINN	MC violation
Fisher-KPP	3e-3 ± 3.5e-4	4.4e-3 ± 7.4e-3	8e-3 ± 0.016	0
Gray-Scott	8.3e-3 ± 3.8e-4	0.011 ± 6.7e-3	0.011 ± 6.7e-3	0
Schnakenberg	9.8e-3 ± 1.8e-4	0.011 ± 2.9e-3	0.011 ± 2.9e-3	0

These cases are not ranked against MC-PINN in the main narrative; on Fisher-KPP Vanilla PINN is lower in L_2 while MC-PINN keeps zero violation; Gray-Scott is snapshot-supervised ($\lambda_r = 0$); on Schnakenberg Fourier-PINN is lowest.

Figure 9 compares Fisher-KPP fields: the front location and shape match, with error concentrated in the front band; MC-PINN keeps zero violation.

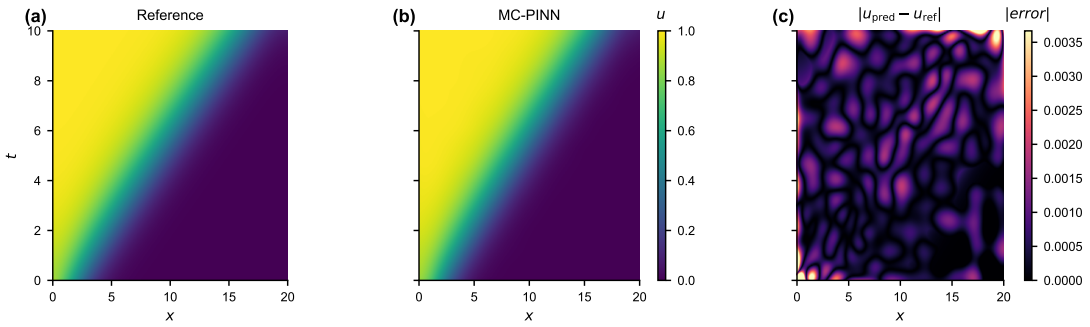


Fig. 9: Fisher-KPP: reference, MC-PINN prediction, and absolute error (seed 20260523). Error concentrates in a narrow front band; flat regions stay near zero; MC-PINN violation is zero.

Mean L_2 does not beat Vanilla PINN (above), so the main gain here is feasibility rather than accuracy ranking.

Gray-Scott is a 2D Turing-pattern case. Figure 10 shows u, v snapshots at IC and late time to test whether bound projection preserves pattern structure under snapshot supervision ($\lambda_r = 0$).

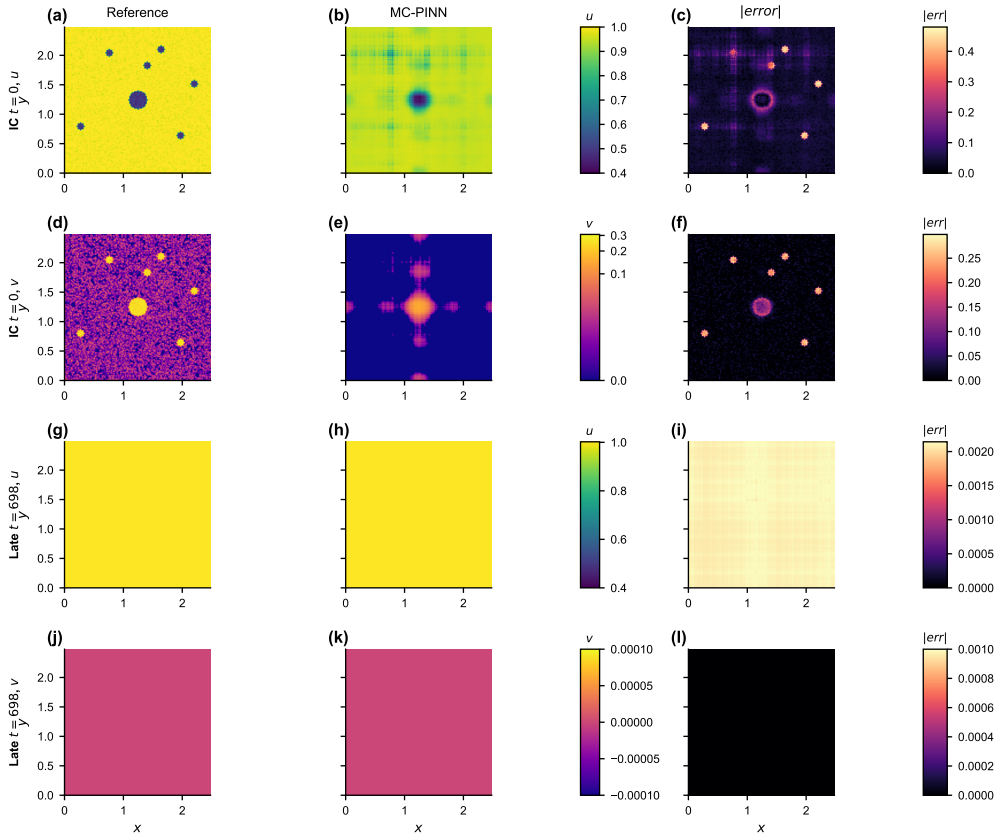


Fig. 10: Gray-Scott: IC and late-time u, v snapshots (reference / MC-PINN / $|error|$). Pattern locations match; error concentrates at edges; IC v uses PowerNorm ($\gamma = 0.28$).

Reference and MC-PINN agree on u, v spatial structure; error concentrates at pattern edges and high-gradient v regions. Fourier-PINN remains lowest in L_2, so this case serves as an applicability control rather than a main-text ranking.

4 Discussion

4.1 Relation to existing PINN routes

MC-PINN is not a universal reaction–diffusion solver but a feasibility-oriented post-projection PINN for stiff heat–species coupling on periodic/zero-flux domains. Unlike Vanilla PINN soft penalties[9,13], we apply alternating box–target-mean projection at inference; unlike PINN-Proj hard conservation[21], the target mean $M_c^\star(t)$ comes from a PDE closure rather than a fixed linear law; unlike Wang et al. causal weighting[17], MC-PINN's staged window serves a stiffness curriculum while the main gain is evaluation-time feasibility correction. Fourier features and spectral informed neural networks mitigate spectral bias and reduce high-order AD cost[16,27], but the L_2 advantage on one-way stiff coupling stems from the projection–constraint bundle, not expressivity alone.

4.2 Main evidence and applicability boundary

Evidence: on one-way stiff coupling, L_2 and constraints improve jointly (Tables 4, 5, Wilcoxon $p = 0.002$, Cohen's $d = 1.06$, lower in 10/10; paired t not significant after Bonferroni); on bidirectional stiff coupling,

MC-PINN keeps near-zero violation and mean residual; bidirectional stiff coupling results appear in Table 6 and Figure 4; on the mild negative control, Vanilla PINN is significantly lower in L_2, showing over-engineering when stiffness is weak. Figures 5–6 show low violation and low projected PDE residuals are not simultaneously achievable—projection trades feasibility for global PDE consistency. Applicability controls in Section 3.9 and Table 9 show MC-PINN gains concentrate on bound/mean feasibility rather than universal L_2 optimality.

4.3 Operator learning and benchmarking culture

Fisher/Gray-Scott/Schnakenberg are not ranked in the main narrative (Section 3.9): different supervision (snapshots, no mean constraint, offline grids)[3,4,12,20]. McGreivy and Hakim warn that weak baselines and reporting bias can inflate operator-learning gains[23]; we therefore treat FNO as complementary only. PDEBench advocates joint accuracy and physics metrics[12]; we further report violation, projection residual, and physics gaps to avoid L_2-only summaries that hide constraint failure.

4.4 Limitations and outlook

Limitations. (1) First-order mean closure makes $M_c^\star(t)$ approximate, with local closure error up to 2.0×10^{-2}; (2) inference-time projection creates a train–inference PDE gap (Table 5); (3) GPU-batched projection reduces overhead to milliseconds but remains post-processing, not differentiable in-training projection; (4) general geometries, multi-species, and non-closure laws need new targets. Future work may explore differentiable approximate projections where closure error is controlled, or embed feasibility layers in operator-learning frameworks[19,22].

5 Conclusion

We proposed MC-PINN, a feasibility-oriented post-projection PINN for stiff heat–species coupling. Ten-seed experiments show Rel. L_2 9.8e-3 on one-way stiff coupling (Wilcoxon $p = 0.002$, Cohen's $d = 1.06$, lower in 10/10) with near-zero violation and projection residual; On bidirectional stiff coupling, MC-PINN keeps near-zero mean residual and violation; on the mild negative control ($k_r = 1$), Vanilla PINN is lower in L_2 (7.9e-3 vs. 0.021), indicating projection is most useful under strong stiffness. Supplementary runs cover strong-constraint baselines, IC-noise robustness, and 2D periodic extensions. The workflow separates the differentiable training field from the projected evaluation field and uses physics gaps together with feasibility–residual Pareto trade-offs to report accuracy and constraint satisfaction jointly; GPU-batched projection adds only millisecond-level overhead as a practical post-processing step. Future work may explore differentiable approximate projections, operator-learning integrations, and closure targets on general geometries and boundary conditions.

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